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Developing device and developing method

Abstract

The developing device comprises: a development chamber that is provided with an air extraction pipeline for extracting air inside the development chamber to outside the development chamber; a carrier that is disposed in the development chamber for supporting a wafer; a plurality of temperature sensors that are disposed on the carrier for detecting temperatures of a plurality of target regions; a plurality of mutually independent air supply pipelines for supplying air to the development chamber, each of the target regions corresponding to at least one air supply pipeline; and a control unit for acquiring measured temperatures of the temperature sensors and calculating current temperatures of the corresponding target regions, and basing on the current temperatures of the target regions to adjust air parameters of the corresponding air supply pipelines, so that the temperatures of the corresponding target regions rest within a preset temperature range.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of International Patent Application No. PCT/CN2021/096008, filed on May 26, 2021, which claims the right of priority to Chinese Patent Application No. 202010510877.0, filed before the Chinese Patent Office on Jun. 8, 2020 and entitled “Developing Device and Developing Method”. The entire contents of International Patent Application No. PCT/CN2021/096008 and Chinese Patent Application No. 202010510877.0 are herein incorporated by reference.

TECHNICAL FIELD
(1) The present application relates to the field of semiconductor fabrication techniques, and more particularly to a developing device and a developing method.
BACKGROUND
(2) In semiconductor fabrication is usually involved a developing operation on photoresist in the manufacturing process. When a wafer is placed on a carrier of the development chamber to be developed there, it is necessary to extract air in real time, so as to timely extract the air produced in the developing process. At present, it is modus operandi to dispose an air supply pipeline right above the wafer and to dispose an air extraction pipeline at the bottom of the development chamber, of which the air supply pipeline is continuously aligned with the wafer to supply air, and the air extraction pipeline continuously extracts the air in the chamber to the outside of the chamber. In the process of air extraction, since the flow rates of the air flowing through various regions on the surface of the wafer are not consistent, evaporation rates at the surface of the wafer are also different, and the temperature at the surface of the wafer changes by gradients, while development rates under different temperatures are different, thereby affecting uniformity of the actual pattern sizes.
SUMMARY
(3) In view of the above technical problem, the present application proposes a developing device and a developing method based on the developing device.
(4) The developing device comprises: a development chamber, provided with an air extraction pipeline for extracting air inside the development chamber to outside the development chamber; a carrier, disposed in the development chamber, for supporting a wafer; a plurality of temperature sensors, disposed on the carrier, for detecting temperatures of a plurality of target regions; a plurality of mutually independent air supply pipelines, for supplying air to the development chamber, each air supply pipeline having a nozzle towards the carrier, each of the target regions corresponding to at least one air supply pipeline; and a control unit, for acquiring measured temperatures of the temperature sensors and basing on the measured temperatures to calculate current temperatures of the corresponding target regions, and basing on the current temperatures of the target regions to adjust air parameters of the corresponding air supply pipelines, so that the temperatures of the corresponding target regions rest within a preset temperature range.
(5) A developing method, which develops on the basis of the aforementioned developing device, and comprises: pouring a developer onto a carrier and placing a wafer surface-coated with photoresist on the carrier; ejecting airflows to surface of the wafer through a plurality of air supply pipelines, and extracting air inside a development chamber to outside the development chamber through an air extraction pipeline in real time; and detecting temperatures of different target regions through temperature sensors on the carrier, acquiring measured temperatures of the temperature sensors and basing on the measured temperatures to calculate current temperatures of corresponding target regions, and basing on the current temperatures of the target regions to adjust

air parameters of the corresponding air supply pipelines, so that the temperatures of the corresponding target regions rest within a preset temperature range.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) In FIG. **1** is a three-dimensional diagram schematically illustrating the structure of the developing device according to an embodiment;

(2) FIG. **2** is a diagram illustrating the relation of electrical connection among the control unit, the temperature sensors and the air supply pipelines according to an embodiment;

(3) FIG. **3** is a three-dimensional diagram schematically illustrating the structure of the developing device with a turntable added according to another embodiment;

(4) FIG. **4** is a diagram illustrating the corresponding relation between target regions and air-blowing regions according to an embodiment;

(5) FIG. **5** is a diagram illustrating the corresponding relation between turntable through-holes and nozzles according to an embodiment;

(6) FIG. **6** is a diagram schematically illustrating the temperature regulator according to an embodiment; and

(7) FIG. **7** is a flowchart illustrating the steps of the developing method according to an embodiment.

NOTES ON REFERENCE NUMERALS

(8) **100**: development chamber **110**: air extraction pipeline **120**: ventilation screen **200**: carrier **210**: temperature sensor **300**: turntable **310**: through-hole **400**: nozzle **410**: temperature regulator **500**: control unit **201**: first target region **202**: second target region **203**: third target region **204**: fourth target region **301**: first air-blowing region **302**: second air-blowing region **303**: third air-blowing region **304**: fourth air-blowing region **411**: pipeline housing **412**: heating module **413**: temperature monitor/control module

DESCRIPTION OF EMBODIMENTS

(9) To facilitate understanding of the present application, the present application will be described more comprehensively below with reference to the accompanying drawings. The accompanying drawings illustrate the most preferred embodiments of the present application. However, the present application can be implemented by many different modes, and are not restricted to the embodiments herein described. To the contrary, the objective for providing these embodiments is to make the contents published by the present application more thorough and comprehensive.

(10) Unless otherwise defined, all technical and scientific terms used throughout the context are identical in meanings as they are usually understood by persons skilled in the art. Such terms as used in the Description of the present application are merely for the purpose of describing specific embodiments, and are not meant to restrict the present application. The wording " . . . and/or . . . " used in this context means the random and all combination(s) of one or more relevantly listed item(s).

(11) As shown in FIG. **1**, the developing device comprises: a development chamber **100**, provided with an air extraction pipeline **110** for extracting air inside the development chamber **100** to outside the development chamber **100**; a carrier **200**, disposed in the development chamber **100**, for supporting a wafer; a plurality of temperature sensors **210**, disposed on the carrier **200**, specifically on an upper surface of the carrier **200**, for detecting temperatures of a plurality of target regions (as shown by dotted lines in FIG. **1**)—when a wafer is placed on the carrier **200**, temperature sensors **210** can detect temperatures of plural target regions of the wafer; a plurality of mutually independent air supply pipelines, for supplying air to the development chamber **100**, each air supply pipeline having a nozzle **400** towards the carrier **200**, air being blown to the surface of the

wafer on the carrier **200** through nozzles **400**, each target region corresponding to at least one nozzle **400** of an air supply pipeline, and temperatures of corresponding target regions being adjustable by changing air parameters of the air supply pipelines. As should be noted, for the sake of clarity, in FIG. **1** are merely shown nozzles **400** of partial air supply pipelines. The shape and distribution of the tubings of the air supply pipelines behind the nozzles **400** can be adjusted as practically required, and no definition is made thereto in this context, as long as it is guaranteed that the nozzles **400** of the various air supply pipelines possess the aforementioned features.

(12) As shown in combination with FIG. **2**, the developing device further comprises a control unit **500** connected to the various temperature sensors **210** and air supply pipelines respectively for acquiring measured temperatures of the temperature sensors **210** and calculating current temperatures of the corresponding target regions, and basing on the current temperatures of the target regions to adjust air parameters of the corresponding air supply pipelines, so that the temperatures of the corresponding target regions rest within a preset temperature range. In one embodiment, when the current temperature of a target region is higher than the preset temperature range, the air parameter of the corresponding air supply pipeline is adjusted to lower the temperature of this target region, until the temperature of this target region is lowered to be within the preset temperature range; when the temperature of a target region is lower than the preset temperature range, the air parameter of the corresponding air supply pipeline is adjusted to raise the temperature of this target region, until the temperature of this target region is raised to be within the preset temperature range. For instance, on the carrier are divided the first to the Nth target regions, each target region is provided with at least one temperature sensor, each target region corresponds to one corresponding set of air supply pipelines, each set of air supply pipelines includes at least one air supply pipeline, the first target region corresponds to the first set of air supply pipelines, the second target region corresponds to the second set of air supply pipelines, so on so forth, and the Nth target region corresponds to the Nth set of air supply pipelines. When the current temperature of the Nth target region exceeds the preset temperature range, the control unit **500** controls the air parameter of the Nth set of air supply pipelines, until the temperature of the Nth target region resumes to be within the preset temperature range.

(13) In the aforementioned developing device, temperature sensors **210** are provided to different target regions on the carrier **200**, when a wafer is placed on the carrier **200**, temperatures of plural regions of the wafer can be monitored and controlled via the temperature sensors **210**. At the same time, plural mutually independent air supply pipelines are disposed above the carrier **200**, nozzles **400** of the air supply pipelines are directed right to the carrier **200** to blow air thereto, each target region corresponds to at least one air supply pipeline, and wafer temperatures of corresponding target regions can be adjusted by adjusting air parameters of the air supply pipelines. The control unit **500** is connected to the various temperature sensors **210** and air supply pipelines respectively for acquiring measured temperatures of the temperature sensors **210** and calculating current temperatures of the target regions on the basis of the measured temperatures, and then basing on the current temperatures of the target regions to adjust air parameters of the corresponding air supply pipelines, so that the measured temperatures of the corresponding temperature sensors **210** resume to be within a preset temperature range. When the current temperature of a target region is higher than the preset temperature range, the air parameter of the corresponding air supply pipeline is adjusted to lower the temperature of this target region, until the temperature of this target region is lowered to be within the preset temperature range; when the temperature of a target region is lower than the preset temperature range, the air parameter of the corresponding air supply pipeline is adjusted to raise the temperature of this target region, until the temperature of this target region is raised to be within the preset temperature range, whereby temperatures of the various regions of the wafer rest only within the preset range, and it is avoided that temperature differences among the various regions of the wafer should be unduly large, thereby ensuring uniformity of etching line widths.

(14) In one embodiment, as shown in FIG. **1**, the air extraction pipeline **110** is disposed at the bottom surface of the development chamber **100**, the carrier **200** is disposed at the bottom of the development chamber **100** and above the air extraction pipeline **110**, when nozzles **400** of the air supply pipelines above the carrier **200** blow air to the carrier **200**, it is ensured that airflow is blown up to down towards the surface of the wafer. In one embodiment, on the top of the development chamber **100** is further provided a ventilation screen **120**. In one embodiment, the various nozzles **400** are directed right to the upper surface of the carrier **200**, the airs ejected out of the nozzles **400** are blown perpendicularly down to the surface of the wafer, whereby is made possible to better adjust temperatures of the corresponding target regions. In one embodiment, the temperature sensors **210** are uniformly distributed on the carrier **200**, each temperature sensor **210** corresponds to one target region, each target region corresponds to a nozzle **400** of one air supply pipeline, and wafer temperatures of the target regions in which the various temperature sensors **210** are located are adjusted through the plurality of air supply pipelines, whereby temperature distribution on the entire surface of the wafer is made more uniform. In other embodiments, in one target region can be provided plural temperature sensors **210**, an average temperature of the plural temperature sensors **210** is calculated to serve as the temperature of the corresponding target region, one temperature region can correspond to the nozzles **400** of plural air supply pipelines, and the temperature of the same and single target region is adjusted through plural nozzles **400** together.
(15) In one embodiment, the aforementioned adjustment of air parameters of the air supply pipelines can be the adjustment of flow rates of airs and temperatures of airs; the greater the flow rate of an air is, the quicker will be the evaporation rate of liquid on the wafer surface, and the lower will be the temperature of the corresponding region; the lower the temperature of an air is, the greater will be the temperature difference between the air and the wafer surface, the quicker will be heat conduction, and the lower will be the wafer temperature of the corresponding region. In this embodiment, it is possible to equalize the flow rates of airs flowing from the nozzles **400** to the wafer surface, while adjustment is only made on the air temperatures. In other embodiments, it is as well possible to only adjust air flow rates or to simultaneously adjust air flow rates and air temperatures.
(16) In one embodiment, as shown in FIGS. **3** and **4**, the carrier **200** has a plurality of target regions in the shape of concentric rings, each target region in the shape of concentric ring is provided therein with a plurality of temperature sensors **210**, and the control unit **500** is employed to acquire measured temperatures of the temperature sensors in the target region and calculate an average temperature of all temperature sensors in the target region to serve as the current temperature of the corresponding target region. In one embodiment, the temperature sensors **210** in a target region in the shape of concentric ring are uniformly distributed. The number of the temperature sensors disposed in a single target region in the shape of concentric ring can be flexibly selected, in this embodiment, the number of the temperature sensors **210** in a target region can be greater than or equal to four—the more the number of temperature sensors **210** is, the more precise will be calculated and obtained the temperature of the target region. The aforementioned developing device further comprises a turntable **300** disposed between the carrier **200** and nozzles **400** of the air supply pipelines, that is to say, the turntable **300** is located above the carrier **200** and below the nozzles **400** of the various air supply pipelines. In one embodiment, the turntable **300** is in a plane that is parallel to the upper surface of the carrier **200**. The turntable **300** has a plurality of air-blowing regions in the shape of concentric rings, the air-blowing regions correspond to the target regions on a one-by-one basis to blow air to corresponding target regions, each air-blowing region in the shape of concentric ring is opened with a plurality of uniformly distributed through-holes **310**, and above each air-blowing region in the shape of concentric ring are provided a plurality of uniformly distributed nozzles **400**. For the sake of clarity, FIG. **3** merely shows nozzles **400** above just one air-blowing region in the shape of concentric ring, as understandable, other air-blowing regions are also provided with nozzles. The turntable **300** is rotatable within a plane parallel to the

upper surface of the carrier **200**, and the through-holes **310** allow airflows of the nozzles **400** to pass during the rotation process. In this embodiment, below the nozzles **400** is provided a turntable **300** with through-holes **310**, and the through-holes **310** are distributed in the shape of concentric rings, thus facilitating control of the airflows of the nozzles to rotate and to be sent to the corresponding target regions in the shape of concentric rings; moreover, flow rates of airflows throughout the same and single target region are essentially the same, and it suffices to adjust the air temperatures in the air supply pipelines during the development process in order to ensure that temperatures of various regions of the wafer are uniform.

(17) The above configuration is exemplarily described with reference to FIG. **4**. The carrier **200** is divided, from the inside out, into a first target region **201**, a second target region **202**, a third target region **203**, and a fourth target region **204**. Of which the first target region **201** is located in a central position of the carrier **200** and assumes a round shape, and the second target region **202**, the third target region **203**, and the fourth target region **204** are respectively assume the shape of concentric rings. The turntable **300** is divided, from the inside out, into a first air-blowing region **301**, a second air-blowing region **302**, a third air-blowing region **303**, and a fourth air-blowing region **304**. The first air-blowing region **301** is located in a central position of the turntable **300** and assumes a round shape, and the second air-blowing region **302**, the third air-blowing region **303**, and the fourth air-blowing region **304** are respectively assume the shape of concentric rings. The first air-blowing region **301** corresponds to the first target region **201**, the second air-blowing region **302** corresponds to the second target region **202**, the third air-blowing region **303** corresponds to the third target region **203**, and the fourth air-blowing region **304** corresponds to the fourth target region **204**. On the turntable **300**, each air-blowing region is opened with a plurality of uniformly distributed through-holes **310**, and above each air-blowing region are provided a plurality of uniformly distributed nozzles **400**. The nozzles above the fourth air-blowing region **304** blow airs to the fourth target region **204** through the through-holes **310** above the fourth air-blowing region, the nozzles **400** above the third air-blowing region **303** blow airs to the third target region **203** through the through-holes **310** above the third air-blowing region **303**, the nozzles **400** above the second air-blowing region **302** blow airs to the second target region **202** through the through-holes **310** above the second air-blowing region **302**, and the nozzles **400** above the first air-blowing region **301** blow airs to the first target region **201** through the through-holes **310** above the first air-blowing region **301**. As should be noted, the aforementioned first to fourth air-blowing regions are merely described by way of example, and the temperature sensors **210** and the through-holes **310** can be respectively formed as more concentric rings. In this embodiment, since the turntable **300** should be rotated during operation, the through-holes **310** on the turntable **300** are distributed in the shape of concentric rings, and the temperature sensors **210** on the carrier **200** are also distributed in the shape of concentric rings, whereby it can be guaranteed that, during the process of rotation, airflows at different positions are allowed to pass when the through-holes **310** are rotated to different positions, and this in turn guarantees that the airflows above the turntable **300** are blown to the wafer surface insofar as possible through the through-holes **310**.

(18) In one embodiment, the various nozzles **400** are within the same plane that is parallel to the upper surface of the carrier **200**, that is to say, the distances between the various nozzles **400** and the turntable **300** are identical, and the turntable **300** is located below the nozzles **400** and abuts against the nozzles **400**, for instance, the distances between the nozzles **400** and the turntable **300** are less than 1 cm. Abutting the nozzles **400** against the turntable **300** makes it possible to guarantee that more airflows ejected from more nozzles **400** pass through the through-holes **310** and arrive at the wafer surface, thus achieving a better control effect.

(19) In one embodiment, as shown in FIGS. **4** and **5**, the air-blowing regions in the shape of concentric rings are opened with the same number of through-holes **310**; in one embodiment, each air-blowing region in the shape of concentric ring is opened with four uniformly distributed through-holes. In one embodiment, each through-hole corresponds to one nozzle **400** of an air

supply pipeline. In other embodiments, the number of through-holes opened in each single air-blowing region may also be a different number, as long as it is guaranteed that the number of through-holes in the single air-blowing region in the shape of concentric ring is greater than or equal to two, whereby is guaranteed that flow rates of airflows supplied to the same target region are identical.

(20) In one embodiment, as shown by a combination of FIG. **4** with FIG. **5**, length of the single through-hole **310** increases with increase of ring diameter of the corresponding air-blowing region in the shape of concentric ring. Since the internal diameter of the concentric ring near the periphery of the turntable **300** is relatively large, the length of a single through-hole **310** is increased, and the through-hole **310** is configured as strip-shaped, then the area of the through-hole **310** can be increased, so that more airflows above the turntable **300** will be blown through the turntable **300** to the wafer surface.

(21) In one embodiment, as shown in FIG. **5**, widths d**1** of the various through-holes **310** are identical. In one embodiment, widths of the through-holes **310** are equal to widths of the corresponding air-blowing regions, that is to say, the through-holes **310** are themselves distributed in the shape of concentric rings, and distances d**2** among adjacent air-blowing regions are identical. In one embodiment, d**1**=d**2**. In one embodiment, 18 mm≤d**1**≤22 mm. In this embodiment, d**1**=20 mm. In one embodiment, as shown in FIG. **5**, numbers of through-holes **310** on the various air-blowing regions in the shape of concentric rings are identical, and lengths of through-holes **310** on the same concentric ring are identical. As should be noted, the numbers and distribution modes of the through-holes **310** are not restricted to what is described above, and the through-holes **310** can also be distributed in other modes in other embodiments. In this embodiment, the mode of distributing the through-holes **310** on the turntable **300** is defined, and this mode can increase the areas of the through-holes **310** as large as possible and enable airflows ejected out of the nozzles **400** to be blown to the wafer surface as far as possible on the one hand, and, on the other hand, also facilitate control of the flow rates of the airs arriving at the same target region of the wafer to be approaching the same during the rotation process of the turntable **300**.

(22) In one embodiment, as shown in FIG. **6**, a temperature regulator **410** is installed in the air supply pipeline, and the control unit **500** specifically changes air temperature in the corresponding air supply pipeline by controlling the temperature of the temperature regulator **410**, so as to adjust the wafer temperature of the region in which a corresponding temperature sensor **210** is located. In one embodiment, the temperature regulator **410** includes a pipeline housing **411**, and a heating module **412** and a temperature monitor/control module **413** in the pipeline, of which the heating module **412** is employed to heat passing airflows (as shown by dotted lines in FIG. **6**), the temperature monitor/control module **413** is employed to monitor and control airflow temperature in the pipeline, the control unit **500** bases on the measured temperature fed back by a temperature sensor **210** to control the heating module **412** to raise or lower the temperature, the heating module **412** controls the airflow temperature of the corresponding air supply pipeline through heat conduction, and the current airflow temperature can be fed back in real time through the temperature monitor/control module **413**.

(23) In the aforementioned developing device, plural temperature sensors **210** are disposed on different regions of the carrier **200** to measure temperatures of different target regions, when a wafer is placed on the carrier **200**, temperatures of the various target regions of the wafer can be monitored and controlled through the temperature sensors **210**. At the same time, plural independent air supply pipelines are provided above the carrier **200**, nozzles **400** of the air supply pipelines blow air right towards the carrier **200**, each target region corresponds to at least one air supply pipeline, and wafer temperatures of the corresponding target regions can be adjusted by adjusting air parameters of the air supply pipelines. The control unit **500** is connected to the various temperature sensors **210** and the various air supply pipelines respectively to acquire measured temperatures of the various temperature sensors **210**, to calculate current temperatures of the target

regions according to the measured temperatures, and then to adjust air parameters of the corresponding air supply pipelines according to the current temperatures of the target regions, so as to enable the temperatures of the corresponding target regions to resume within a preset temperature range. When the current temperature of a target region is higher than the preset temperature range, the air parameter of the corresponding air supply pipeline is adjusted to lower the temperature of this target region, until the temperature of this target region is lowered to be within the preset temperature range; when the temperature of a target region is lower than the preset temperature range, the air parameter of the corresponding air supply pipeline is adjusted to raise the temperature of this target region, until the temperature of this target region is raised to be within the preset temperature range, whereby is guaranteed that temperatures of the various regions of the wafer rest only within the preset range, and it is avoided that temperature differences among the various regions of the wafer should be unduly large, thereby ensuring uniformity of etching line widths.

(24) The present application further provides a developing method based on the aforementioned developing device, as shown in FIG. **7**, the developing method comprises the following steps.

(25) Step S**100**: pouring a developer onto a carrier and placing a wafer surface-coated with photoresist on the carrier.

(26) Step S**200**: ejecting airflows to surface of the wafer through a plurality of air supply pipelines, and extracting air inside a development chamber to outside the development chamber through an air extraction pipeline in real time.

(27) In one embodiment, when airflows are ejected to the surface of the wafer through a plurality of air supply pipelines, the airflow flow rates of the various air supply pipelines are so adjusted that air speed is identical throughout the wafer surface.

(28) In one embodiment, between the carrier **200** and the nozzles **400** is further disposed a turntable **300**, and the turntable **300** are provided thereon with through-holes **310** allowing airflows of the nozzles **400** to pass. During air extraction, the turntable **300** is controlled to be rotated within a plane parallel to the upper surface of the carrier **200**, on the one hand, airflows of the nozzles **400** can be blown to the wafer surface through the through-holes **310**, and on the other hand, air speed can be controlled to be identical throughout the various regions on the wafer surface by controlling the turntable **300** to be rotated.

(29) Step S**300**: detecting temperatures of different target regions through temperature sensors on the carrier, acquiring measured temperatures of the temperature sensors and basing on the measured temperatures to calculate current temperatures of corresponding target regions, and basing on the current temperatures of the target regions to adjust air parameters of the corresponding air supply pipelines, so that the temperatures of the corresponding target regions rest within a preset temperature range.

(30) In one embodiment, the aforementioned adjustment of air parameters of the air supply pipelines can specifically be the adjustment of flow rates of airs and temperatures of airs; the greater the flow rate of an air is, the quicker will be the evaporation rate of liquid on the wafer surface, and the lower will be the temperature of the corresponding region of the wafer; the lower the temperature of an air is, the greater will be the temperature difference between the air and the wafer surface, the quicker will be heat conduction, and the lower will be the wafer temperature of the corresponding region. In this embodiment, it is possible to equalize the flow rates of airs flowing from the nozzles **400** to the wafer surface, while adjustment is only made on the air temperatures to adjust the temperature of the wafer surface, i.e., the step of ejecting airflows to surface of the wafer through a plurality of air supply pipelines includes adjusting airflow flow rates of the air supply pipelines, so that air speed is identical throughout the surface of the wafer; and the step of basing on the measured temperatures to calculate current temperatures of corresponding target regions, and basing on the current temperatures of the target regions to adjust air parameters of the corresponding air supply pipelines, so that the temperatures of the corresponding target

regions rest within a preset temperature range includes basing on the current temperatures of the target regions to adjust air temperatures of the corresponding air supply pipelines, so that the temperatures of the corresponding target regions rest within a preset temperature range. In other embodiments, it is as well possible to only adjust air flow rates or to simultaneously adjust air flow rates and air temperatures. In one embodiment, a temperature regulator **410** is installed in the air supply pipeline, and the above adjustment changes air temperature in the corresponding air supply pipeline by controlling the temperature of the temperature regulator **410**, so as to adjust the wafer temperature of the region in which a corresponding temperature sensor **210** is located.
(31) In one embodiment, the step of equalizing the flow rates of airs arriving at the various regions on the wafer surface and adjusting air parameters of the corresponding air supply pipelines so that the temperatures of the corresponding target regions rest within a preset temperature range includes basing on the current temperatures of the target regions to adjust air temperatures of the corresponding air supply pipelines so that the temperatures of the corresponding target regions rest within a preset temperature range.
(32) As should be understood, although the various steps shown in the flowchart of FIG. **7** are sequentially illustrated according to arrow indications, these steps are NOT necessarily executed in the sequence according to the arrow indications. Unless explicitly explained in this context, the execution of these steps does not have to abide by a strict sequence, as they can also be executed by other sequences. Moreover, at least partial steps in FIG. **7** can include plural sub-steps or plural sub-phases, these sub-steps or sub-phases are not necessarily executed at the same timing, but they can be executed at different timings, and their execution is not necessarily according to sequence, as they can be executed in turns or alternatively with respect to at least part of other steps or sub-steps or sub-phases of other steps.
(33) In the aforementioned developing method, plural temperature sensors **210** are disposed on different regions of the carrier **200** to measure temperatures of different target regions, when a wafer is placed on the carrier **200**, temperatures of the various target regions of the wafer can be monitored and controlled through the temperature sensors **210**. At the same time, plural independent air supply pipelines are provided above the carrier **200**, nozzles **400** of the air supply pipelines blow air right towards the carrier **200**, each target region corresponds to at least one air supply pipeline, and wafer temperatures of the corresponding target regions can be adjusted by adjusting air parameters of the air supply pipelines. The control unit **500** is connected to the various temperature sensors **210** and the various air supply pipelines respectively to acquire measured temperatures of the various temperature sensors **210**, to calculate current temperatures of the target regions according to the measured temperatures, and then to adjust air parameters of the corresponding air supply pipelines according to the current temperatures of the target regions, so as to enable the temperatures of the corresponding target regions to resume within a preset temperature range. When the current temperature of a target region is higher than the preset temperature range, the air parameter of the corresponding air supply pipeline is adjusted to lower the temperature of this target region, until the temperature of this target region is lowered to be within the preset temperature range; when the temperature of a target region is lower than the preset temperature range, the air parameter of the corresponding air supply pipeline is adjusted to raise the temperature of this target region, until the temperature of this target region is raised to be within the preset temperature range, whereby is guaranteed that temperatures of the various regions of the wafer rest only within the preset range, and it is avoided that temperature differences among the various regions of the wafer should be unduly large, thereby ensuring uniformity of etching line widths.
(34) The foregoing embodiments merely represent several modes of execution of the present application, they are described relatively in specifics and in detail, but they should not be therefore understood as restricting the scope of the present application. As should be noted, persons ordinarily skilled in the art may make various modifications and improvements without departing

from the conception of the present application, and all such modifications and improvements shall fall within the protection scope of the present application. Accordingly, the protection scope of the present application shall base on the attached Claims.

Claims

1. A developing device, characterized in comprising: a development chamber, provided with an air extraction pipeline for extracting air inside the development chamber to outside the development chamber; a carrier, disposed in the development chamber, for supporting a wafer; a plurality of temperature sensors, disposed on the carrier, for detecting temperatures of a plurality of target regions; a plurality of mutually independent air supply pipelines, for supplying air to the development chamber, each air supply pipeline having a nozzle towards the carrier, each of the target regions corresponding to at least one air supply pipeline; and a control unit, for acquiring measured temperatures of the temperature sensors and basing on the measured temperatures to calculate current temperatures of the corresponding target regions, and basing on the current temperatures of the target regions to adjust air parameters of the corresponding air supply pipelines, so that the temperatures of the corresponding target regions rest within a preset temperature range; wherein the carrier has a plurality of target regions in the shape of concentric rings, each target region in the shape of concentric ring is provided therein with a plurality of uniformly distributed temperature sensors, and the control unit is employed to acquire measured temperatures of the temperature sensors in the target region and calculate an average temperature of all temperature sensors in the target region to serve as the current temperature of the corresponding target region; the developing device further comprising: a turntable, disposed between the carrier and all nozzles, the turntable having a plurality of air-blowing regions in the shape of concentric rings, the air-blowing regions corresponding to the target regions on a one-by-one basis to blow air to corresponding target regions, each air-blowing region in the shape of concentric ring being opened with a plurality of uniformly distributed through-holes, above each air-blowing region in the shape of concentric ring being provided a plurality of uniformly distributed nozzles, the turntable being rotatable within a plane parallel to the carrier, and the through-holes allowing airflows of the nozzles to pass during rotation process.
2. The developing device according to claim 1, wherein the nozzles are within the same and single plane parallel to an upper surface of the carrier, and the turntable is located below and abuts against the nozzles.
3. The developing device according to claim 1, wherein the air-blowing regions in the shape of concentric rings are opened with the same number of through-holes.
4. The developing device according to claim 3, wherein each air-blowing region in the shape of concentric ring is opened with four uniformly distributed through-holes, and each through-hole corresponds to one nozzle.
5. The developing device according to claim 3, wherein length of a single through-hole increases with increase of ring diameter of the corresponding air-blowing region in the shape of concentric ring.
6. The developing device according to claim 1, wherein widths of the through-holes opened in the air-blowing regions are equal to widths of the corresponding air-blowing regions.
7. The developing device according to claim 6, wherein the adjacent air-blowing regions are arranged with an interval there between, and the intervals amongst all the adjacent air-blowing regions are identical.
8. The developing device according to claim 4, wherein intervals amongst the adjacent air-blowing regions are equal to widths of the through-holes.
9. The developing device according to claim 1 wherein a temperature regulator is installed in the

air supply pipeline, and the control unit changes air temperature in the corresponding air supply pipeline by controlling the temperature regulator.